State Estimation and Filtering for Robotic Perception

AA273

Kalman Filter Derivation



Stanford University

MSL

AA273: State Estimation and Filtering for Robotic Perception

Announcements:

- HW 3 due Fri Jan 30
- HW 4 out Fri Jan 30, due Fri Feb 13
- Group signup sheet posted (groups from 1 – 4 people)
- Midterm: Feb 5-7
  - 5 hour timed take home exam, you choose any 5 hours in 48 hour window
  - Open book, notes, lecture videos
  - No internet, gen AI, or collaboration, coding assistants are allowed.

Lecture 7:

- Kalman Filter derivation

Recall: Linear-Gaussian States Space Model

Dynamics eqn.: $X_{t+1} = A_t X_t + B_t u_t + W_t \qquad W_t \sim GWN(0, Q_t)$

Measurement eqn.: $Y_t = C_t X_t + D_t u_t + V_t \qquad V_t \sim GWN(0, R_t)$

Initial condition: $X_0 \sim \mathcal{N}(\mu_{0|0}, \Sigma_{0|0})$ $\qquad X_0, W_t, V_\tau$ independent for all t, τ

Claim: posterior $p(x_t \mid y_{1:t})$ and prediction $p(x_{t+1} \mid y_{1:t})$ is Gaussian for all t.

Notation: $p(x_t \mid y_{1:t}) = \mathcal{N}(\mu_{t|t}, \Sigma_{t|t})$

$$p(x_{t+1} \mid y_{1:t}) = \mathcal{N}(\mu_{t+1|t}, \Sigma_{t+1|t})$$

We compute these with the Kalman Filter

Recall: Kalman Filter Equations

Given: $\mu_{0|0}, \Sigma_{0|0},\ A_t, B_t, C_t,\ Q_t, R_t$

(Assume $D_t = 0$)

For: $t = 1, 2, \ldots$

Prediction:

$$\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}$$

$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$$

Update:

$$\mu_{t|t} = \mu_{t|t-1} + K_t(y_t - C_t\mu_{t|t-1})$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_tC_t\Sigma_{t|t-1}$$

$$K_t = \Sigma_{t|t-1}C_t^T(C_t\Sigma_{t|t-1}C_t^T + R_t)^{-1}$$

Loop

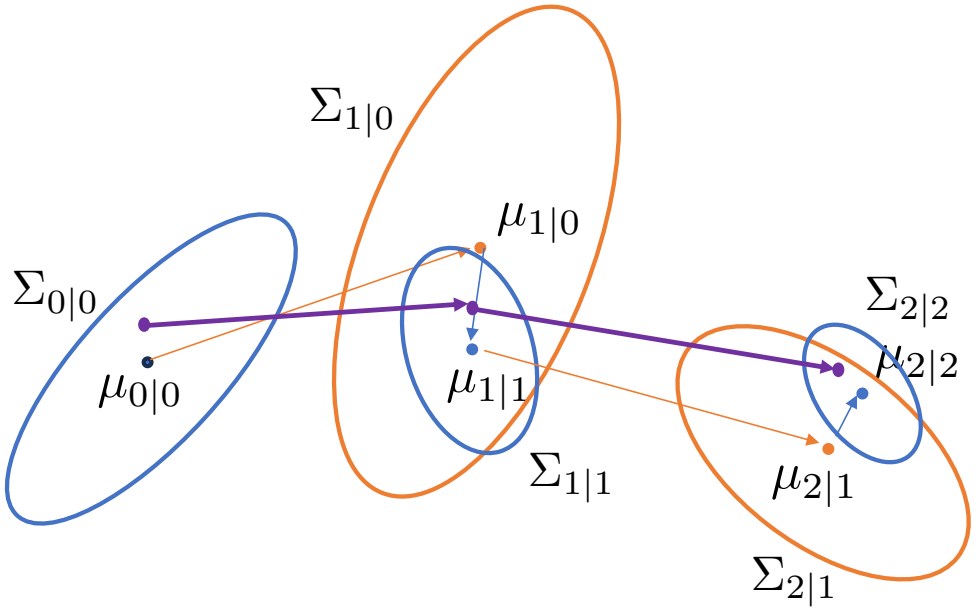
Compare update with Gaussian Est. Eqns:

$$\mu_{x|y} = \mu_x + \Sigma_{xy}\Sigma_y^{-1}(y - \mu_y)$$

$$\Sigma_{x|y} = \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}$$

Notice the same form!

KF Illustration



(i) Predict step increases uncertainty

$$\Sigma_{t|t-1} \geq \Sigma_{t-1|t-1}$$

(usually)

(ii) Update step decreases uncertainty

$$\Sigma_{t|t} \leq \Sigma_{t|t-1}$$

KF Derivation, Update Step

Rewrite Gaussian estimation equations with KF notation:

$$\mu_{t|t} = \mu_{t|t-1} + \Sigma_{x_t y_t} \Sigma_{y_t}^{-1} (y_t - \mu_{y_t})$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{x_t y_t} \Sigma_{y_t}^{-1} \Sigma_{y_t x_t}$$

Where:

$$\mu_{y_t} = E[Y_t \mid Y_{1:t-1}]$$

$$\Sigma_{x_t y_t} = E[(X_t - \mu_{t|t-1})(Y_t - \mu_{y_t})^T \mid Y_{1:t-1}]$$

$$\Sigma_{y_t} = E[(Y_t - \mu_{y_t})(Y_t - \mu_{y_t})^T \mid Y_{1:t-1}]$$

Recall SS model:

$$X_{t+1} = A_t X_t + B_t u_t + W_t$$

$$Y_t = C_t X_t + D_t u_t + V_t$$

KF Derivation Measurement Mean

$$\mu_{y_t} = E[Y_t \mid Y_{1:t-1}]$$

Recall SS measurement eqn: $Y_t = C_t X_t + D_t u_t + V_t$

$$= E[C_t X_t + D_t u_t + V_t \mid Y_{1:t-1}]$$

$$= E[C_t X_t \mid Y_{1:t-1}] + E[D_t u_t \mid Y_{1:t-1}] + E[V_t \mid Y_{1:t-1}]$$

$$= C_t \underbrace{E[X_t \mid Y_{1:t-1}]}_{\mu_{t|t-1}} + \underbrace{D_t u_t E[1 \mid Y_{1:t-1}]}_{=0} + \underbrace{E[V_t \mid Y_{1:t-1}]}_{=E[V_t]=0}$$

A0: Assume $D_t = 0$

A1: Assume V_t independent of $Y_{1:t-1}$

$$\boxed{\mu_{y_t} = C_t \mu_{t|t-1}}$$

We'll prove later A1 follows from SS whiteness assumptions.

KF Derivation State-Measurement Cross Covariance

$$\Sigma_{x_t y_t} = E[(X_t - \mu_{t|t-1})(Y_t - \mu_{y_t})^T \mid Y_{1:t-1}]$$

Recall SS measurement eqn: $Y_t = C_t X_t + D_t u_t + V_t$

Derivation on board and in notes.

$$\Sigma_{x_t y_t} = \Sigma_{t|t-1} C_t^T$$

A2: Assume X_t independent of V_t given $Y_{1:t-1}$

KF Derivation Measurement Covariance

$$\Sigma_{y_t} = E[(Y_t - \mu_{y_t})(Y_t - \mu_{y_t})^T \mid Y_{1:t-1}]$$

Recall SS model: $Y_t = C_t X_t + D_t u_t + V_t$

Derivation on board and in notes.

$$\boxed{\Sigma_{y_t} = C_t \Sigma_{t|t-1} C_t^T + R_t}$$

Uses A1 and A2.

KF Derivation Collecting Results

$$\mu_{y_t} = C_t \mu_{t|t-1}$$
$$\Sigma_{x_t y_t} = \Sigma_{t|t-1} C_t^T$$
$$\Sigma_{y_t} = C_t \Sigma_{t|t-1} C_t^T + R_t$$

True under A0, A1, and A2.

$$\mu_{t|t} = \mu_{t|t-1} + \Sigma_{x_t y_t} \Sigma_{y_t}^{-1} (y_t - \mu_{y_t})$$
$$\mu_{t|t} = \mu_{t|t-1} + \underbrace{(\Sigma_{t|t-1} C_t^T (C_t \Sigma_{t|t-1} C_t^T + R_t)^{-1})}_{K_t} (y_t - C_t \mu_{t|t-1})$$
$$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{x_t y_t} \Sigma_{y_t}^{-1} \Sigma_{y_t x_t}$$
$$\mu_{t|t} = \mu_{t|t-1} + \underbrace{(\Sigma_{t|t-1} C_t^T (C_t \Sigma_{t|t-1} C_t^T + R_t)^{-1})}_{K_t} C_t \Sigma_{t|t-1}$$

$$\mu_{t|t} = \mu_{t|t-1} + K_t (y_t - C_t \mu_{t|t-1})$$
$$\Sigma_{t|t} = \Sigma_{t|t-1} - K_t C_t \Sigma_{t|t-1}$$
$$K_t = \Sigma_{t|t-1} C_t^T (C_t \Sigma_{t|t-1} C_t^T + R_t)^{-1}$$

As desired!

KF Derivation, Covariance Update Step

$$\mu_{t|t-1} := E[X_t \mid Y_{1:t-1}]$$

Recall SS model: $X_t = A_{t-1}X_{t-1} + B_{t-1}u_{t-1} + W_{t-1}$

$$= E[A_{t-1}X_{t-1} + B_{t-1}u_{t-1} + W_{t-1} \mid Y_{1:t-1}]$$

$$= \underbrace{E[A_{t-1}X_{t-1} \mid Y_{1:t-1}]}_{A_{t-1}\mu_{t-1|t-1} \text{ (By def.)}} + \underbrace{E[B_{t-1}u_{t-1} \mid Y_{1:t-1}]}_{B_{t-1}u_{t-1} \text{ (Not random)}} + \underbrace{E[W_{t-1} \mid Y_{1:t-1}]}_{= E[W_{t-1}] = 0}$$

$$\boxed{\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}}$$

As desired

Requires new assumption:

A3: W_t indep. of $Y_{1:t}$

KF Derivation, Mean Prediction Step

$$\mu_{t|t-1} := E[X_t \mid Y_{1:t-1}]$$

Recall SS model: $X_t = A_{t-1}X_{t-1} + B_{t-1}u_{t-1} + W_{t-1}$

$$= E[A_{t-1}X_{t-1} + B_{t-1}u_{t-1} + W_{t-1} \mid Y_{1:t-1}]$$

$$= \underbrace{E[A_{t-1}X_{t-1} \mid Y_{1:t-1}]}_{A_{t-1}\mu_{t-1|t-1} \text{ (By def.)}} + \underbrace{E[B_{t-1}u_{t-1} \mid Y_{1:t-1}]}_{B_{t-1}u_{t-1} \text{ (Not random)}} + \underbrace{E[W_{t-1} \mid Y_{1:t-1}]}_{= E[W_{t-1}] = 0}$$

$$\boxed{\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}}$$

As desired

Requires new assumption:

A3: W_t indep. of $Y_{1:t}$

KF Derivation, Covariance Prediction Step

$$\Sigma_{t|t-1} := E[(X_t - \mu_{t|t-1})(X_t - \mu_{t|t-1})^T \mid Y_{1:t-1}]$$

Recall SS model: $X_t = A_{t-1}X_{t-1} + B_{t-1}u_{t-1} + W_{t-1}$

From previous slide: $\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}$

Derivation on board and in notes.

$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$$

As desired

Requires new assumption:

A4: W_t indep. of X_t given $Y_{1:t}$

KF Derivation, Summarize Prediction Step

$$\mu_{t|t-1} = A_{t-1}\mu_{t-1|t-1} + B_{t-1}u_{t-1}$$
$$\Sigma_{t|t-1} = A_{t-1}\Sigma_{t-1|t-1}A_{t-1}^T + Q_{t-1}$$

As desired!

True under A3 and A4.

Key Assumptions

A0: Assume $D_t = 0$

Straightforward to relax KF to allow for $D_t \neq 0$

A1: Assume V_t indep. of $Y_{1:t-1}$

A2: Assume X_t indep, of V_t given $Y_{1:t-1}$

A3: Assume W_t indep. of $Y_{1:t}$

A4: Assume W_t indep. of X_t given $Y_{1:t}$

These all follow from L-G SS model def:

$$W_t \sim GWN(0, Q_t)$$
$$V_t \sim GWN(0, R_t)$$
$$X_0, W_t, V_\tau \quad \text{independent for all} \quad t, \tau$$

We'll prove next time!